

TrES-3b

R-1824

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150512 · created 2026-08-03T05:03:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457154.8745 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.043
Transit depth (Rp/Rs)^2	3.73 +/- 1.68 [%]
Orbital Inclination (inc)	80.99 +/- 1.89
Airmass coefficient 1 (a1)	1.006 +/- 0.017
Airmass coefficient 2 (a2)	-0.004 +/- 0.04
Scatter in the residuals of the lightcurve fit is	1.66 %
Best Comparison Star	#1 - [481, 371]
Optimal Aperture	4.84
Optimal Annulus	10.07
Transit Duration (day)	0.044 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.043 vs 0.16309 (deviation 0.7σ)
Duration fit vs published	0.044 d vs 0.0567 d (deviation 0.53σ)
Mid-time O–C	2.1 min
Depth SNR	4.5
Residual scatter	1.66 %

Light curve

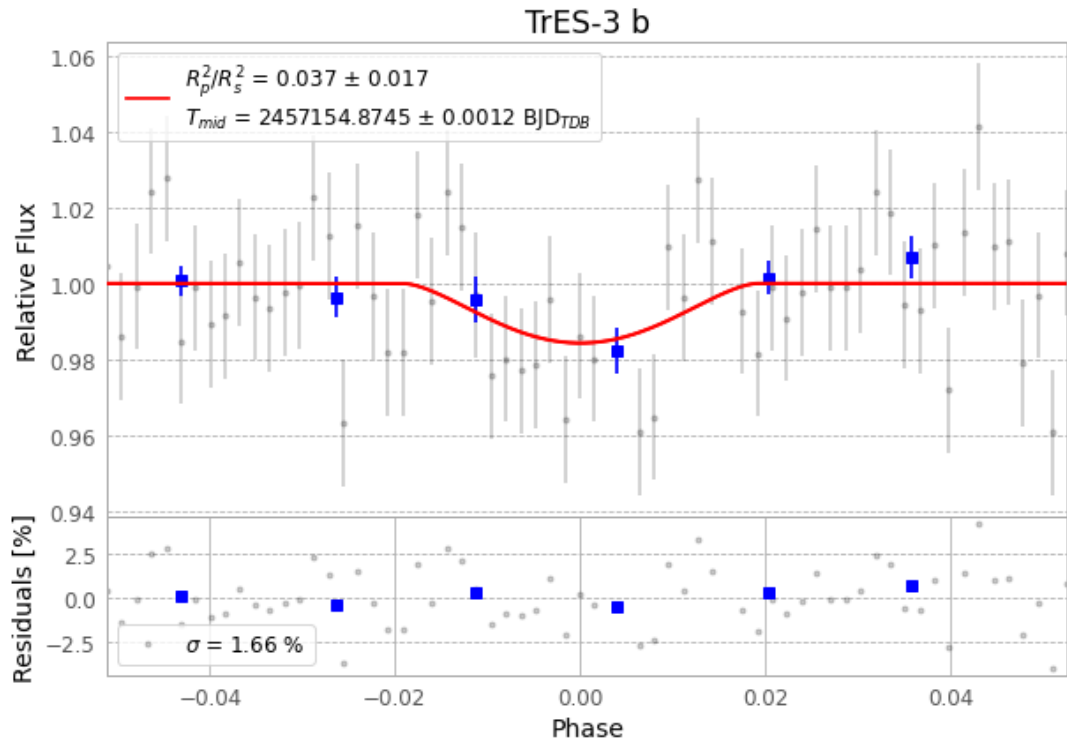


Figure 1 — FinalLightCurve_TrES-3 b_2015-05-12.png (SHA-256 2691211b52b73973...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 442c1b9fba1f8a98...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3150512071812.FITS → TRES-3150512103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150512.log (d72b4081b92a...), FinalLightCurve_TrES-3 b_2015-05-12.png (2691211b52b7...), FinalLightCurve_TrES-3 b_2015-05-12.csv (d6d51e3a7faf...)

Compute resources

Wall time 185.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:03Z.